Same score, different misses

Day 04: keyword search vs embeddings on the same 44 chunks.

Q: How much time off do new dads get?

ab BM25 · MATCHES WORDS

top 3, and the one word each shares:

#1	#what-not-to-do	"off"
#2	#purpose	"time"
#3	#your-first-day	"new"

sections of 3 unrelated policies

⊗ No chunk says "dads".
The answer never reaches the top 3.

EMBEDDINGS · MEANING

top 3, cosine similarity:

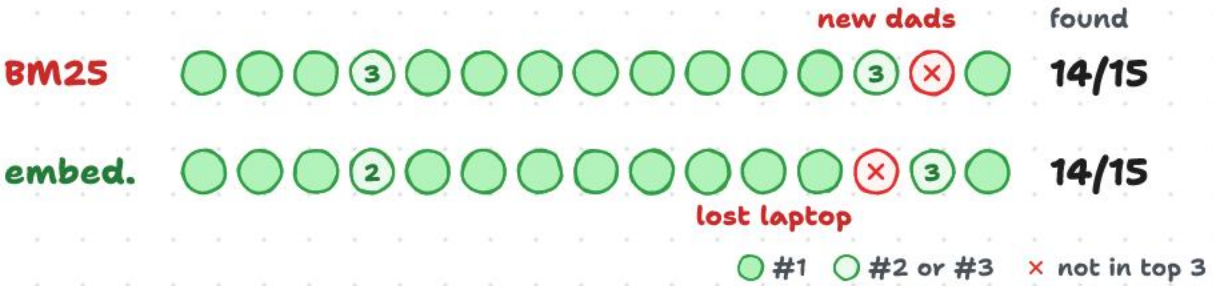
#1	#returning-to-work	0.72
#2	#who-is-eligible	0.67
#3	#leave-entitlement	0.66

"Primary caregivers receive 16 weeks..."

✓ Right policy. Answer at #3.
All three: parental-leave-policy

15 answerable questions · top 3 · same chunks

Where is the chunk with the complete evidence?



Different misses: together they find 15/15.
That's the case for hybrid search (Day 08).

Real run · Day 04 notebook · local model, free, no API key

Text in, 384 numbers out

bge-small-en-v1.5 · MIT · 33M parameters · runs on a CPU

1 **TOKENS**

`[CLS] how much time off do new dad ##s get ? [SEP]`

2 **MODEL**

12 tokens in → 12 vectors of 384 numbers out

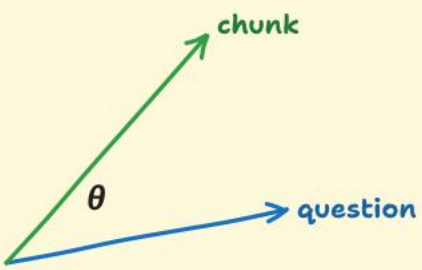
3 **POOL**

keep the [CLS] vector: this model's rule (others average)

4 **NORMALISE**

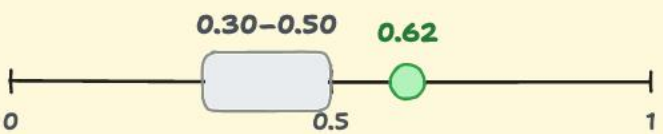
divide by its length → every vector has length 1

COMPARE: COSINE SIMILARITY



length 1 → cosine = dot product

Scores are relative, not %



Gray: 9 pairs of unrelated sentences. Green: "new dads" vs "secondary caregivers".

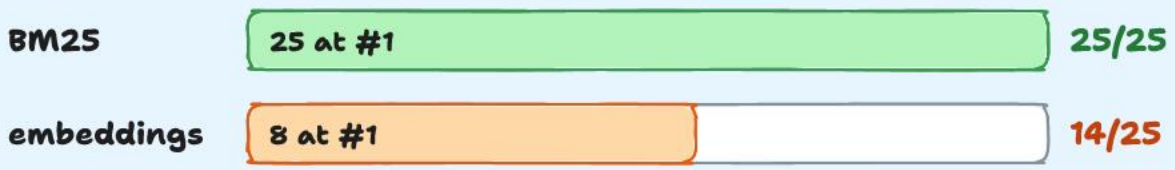
Rank by score. It's not a percentage.

Where each one fails

Two systematic tests on the same chunks, top 3.

EXACT VALUES · BM25'S HOME GROUND

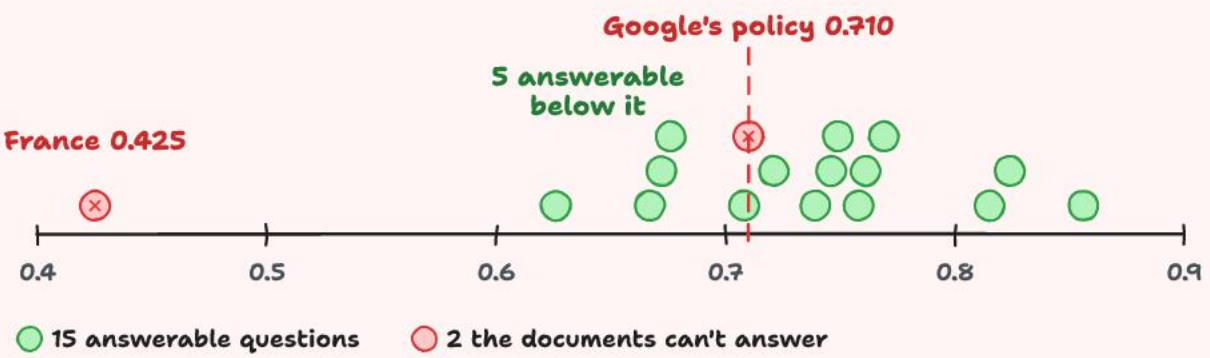
Every number in the documents as a query: 25 (USD 220, 13 months, 15:00...)



"USD 220" (Singapore hotel) → embeddings return chunks with USD 75, USD 500 and USD 1,000. One amount looks like another.

NO SUCH THING AS "NO MATCH"

Embeddings always return 3. Top-1 cosine per question:



A threshold that rejects Google's also rejects Singapore's hotel limit (0.708).

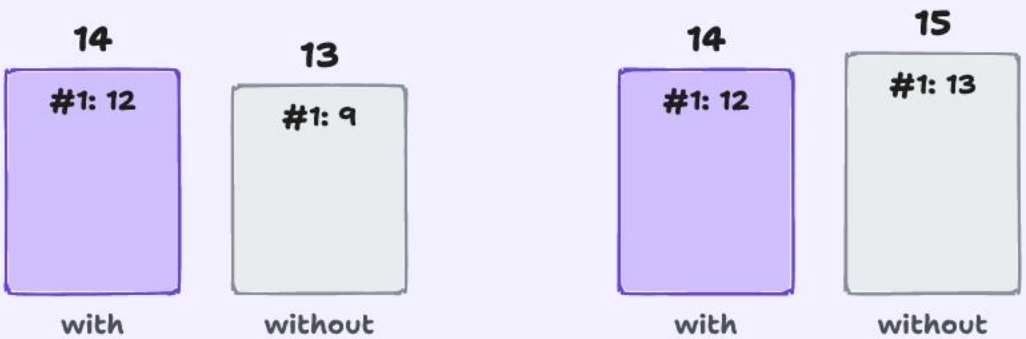
BM25: France → 0 hits, abstains. Google → 6.21, doesn't.

Similarity means "same topic", not "answers the question".

Same knob, opposite effects

The "Title > Section" header line, on 15 answerable questions.

FOUND IN THE TOP 3 (AND AT #1)



Why: a shared header pulls a document's chunks together (similarity 0.65 → 0.71). "New dads" drops from #1 to #3.

A VECTOR BELONGS TO ITS MODEL

model	BAAI/bge-small-en-v1.5
revision	5c38ec7c405e…
pooling	CLS, length 1
chunk_hash	only changed chunks
allowed_groups	travels with it

Edit 1 section: 1 of 44 re-embedded.

New model or revision: re-embed all 44.

Pinned commit + SHA-256. ONNX, no pickle.

Model 133 MB, once. Then it runs offline.

Next → Day 05: local vector search and data lifecycle